

processing.org/download/?processing

- Extract it to any folder of your choice.
- Download the Android SDK from: <http://developer.android.com/sdk/index.html>. Extract it where ever you like.
- Go into the extracted Android-SDK folder, into the tools subfolder and run *android*. In Linux, just open the terminal, *cd* to that folder and type *./android*
- Through the GUI that launches, update and install all available packages.
- The main package is the Android SDK *Platform-tools*
- You can install the version of Android you want. Since I had a phone with Android 2.3, that is what I installed.
- Windows users need the Google USB drivers too.
- Once this is done, your system is ready to develop Android apps.

Coding

- On the terminal, *cd* to the folder in which you have extracted Processing and type *./processing*. Make sure that the file *Processing* has the executable permission set.
- Start Processing. On the top right corner you will find a box called *standard*. Click on it and select *android*.
- Now write down the code. Most of the code in Processing can be used in Android, but there are certain exceptions.
- Here's a small code example, in which we are creating an app to use a finger to draw on the smartphone screen.

```
void setup()
{
    size(640,360);
    background(100);
    smooth();
    stroke(255);
}

void draw()
{
    if(mousePressed==true)
    {
        ellipse(mouseX,mouseY,5,5);
    }
}
```

- Press the *Play* button on the top left of the Processing window and wait till the emulator is loaded with the app.
- You can also connect your Android device to your PC and make the app run on your device. For that, you will have to activate the USB debugging mode in your Android device.
- Once you have tested your sketch and are satisfied, it is time to distribute it to the world and for that we have to package it.

Generating the .apk file

- From the file menu in Processing, select *Export Android Project*. A folder called *Android* is created in your sketch folder.



Figure 1: Android mode

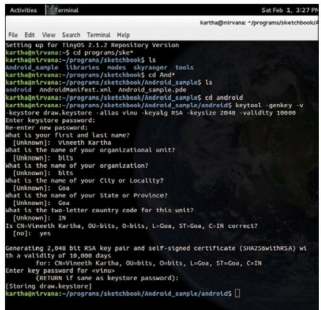


Figure 2: Generate the keystore

- Open the terminal and *cd* to that particular folder.
- Then generate a private-public keypair, as follows:

```
$ keytool -genkey -v -keystore <keyfile.keystore> -alias <alias> -keyalg RSA -keysize 2048 -validity 10000
```

This command is nothing but generating a key and calling it *draw.keystore*. And it is valid for 10,000 days.

Answer all the questions which follow and remember those answers.

- Next, build the *.apk*:

```
$ ant release
```

This command will generate a file called *sketchname-release-unsigned.apk* in the bin folder.

- Sign the app:

```
$ jarsigner -verbose -keystore <keyfile.keystore> bin/draw-
```